

Exp-7:

Read captured video in python and display the video, in slow motion and in fast motion.

Code:

```
import cv2

video = cv2.VideoCapture("sample_video.mp4")

print("Press 'q' to exit")

while video.isOpened():

    ret, frame = video.read()

    if not ret:

        break

    cv2.imshow("Normal Video", frame)

    cv2.imshow("Slow Motion", frame)

    cv2.waitKey(100)

    for i in range(2):

        video.read()

        cv2.imshow("Fast Motion", frame)

        cv2.waitKey(10)

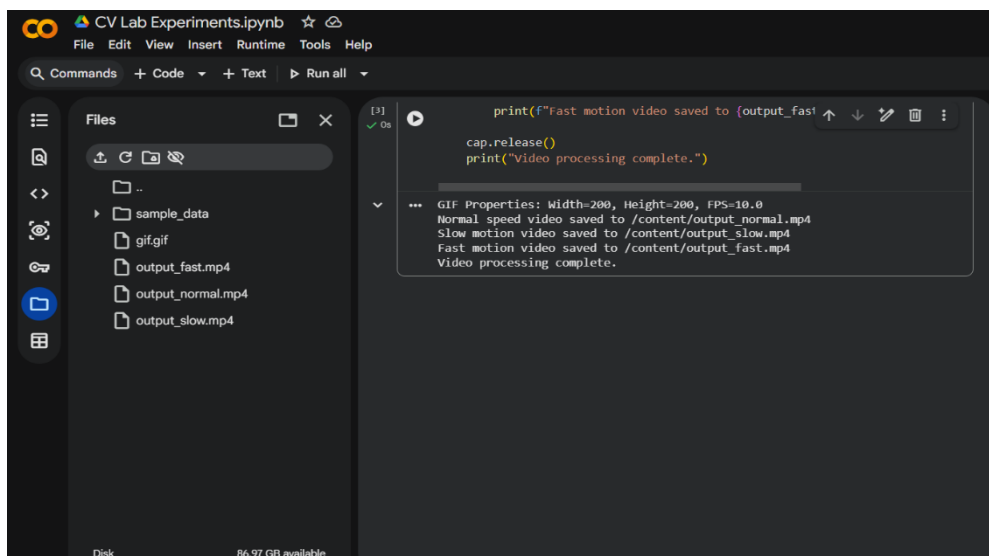
    if cv2.waitKey(1) & 0xFF == ord('q'):

        break

video.release()

cv2.destroyAllWindows()
```

Output:



Exp-8:

Image Handling and processing operations

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab.patches import cv2_imshow
img = cv2.imread("/content/cvexp.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = np.ones((5,5), np.uint8)
dilated = cv2.dilate(gray, kernel, iterations=1)
plt.figure(figsize=(15, 5))
plt.subplot(1, 3, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis('off')
plt.subplot(1, 3, 2)
plt.imshow(gray, cmap='gray')
plt.title("Grayscale Image")
plt.axis('off')
plt.subplot(1, 3, 3)
plt.imshow(dilated, cmap='gray')
plt.title("Dilated Image")
plt.axis('off')
plt.tight_layout()
plt.show()
```

Output:



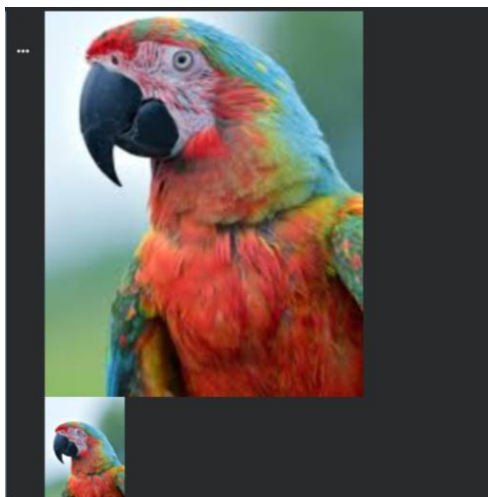
Exp-9:

Image scaling techniques to resize the images to both bigger and small sizes.

Code:

```
import cv2
img = cv2.imread("input.jpg")
height, width = img.shape[:2]
bigger = cv2.resize(img, None, fx=2, fy=2, interpolation=cv2.INTER_LINEAR)
smaller = cv2.resize(img, None, fx=0.5, fy=0.5, interpolation=cv2.INTER_AREA)
cv2.imshow("Original Image", img)
cv2.imshow("Scaled Up Image (2x)", bigger)
cv2.imshow("Scaled Down Image (0.5x)", smaller)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:



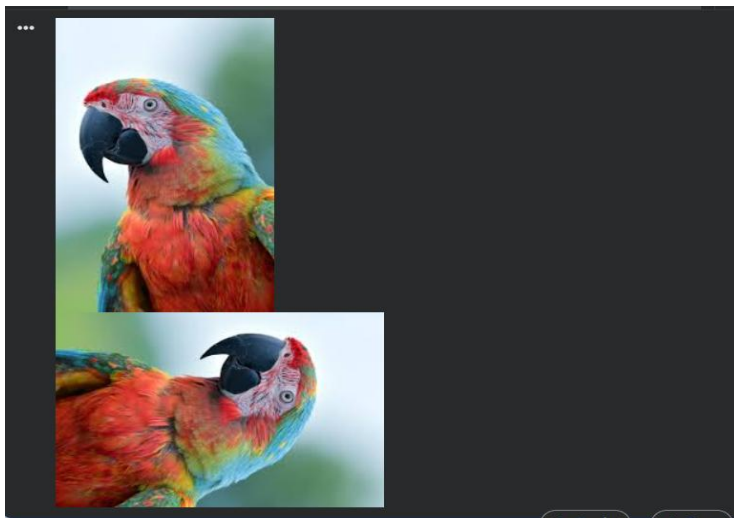
Exp-10:

90-degree rotation clockwise

Code:

```
import cv2  
img = cv2.imread("input.jpg")  
rotated = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)  
cv2.imshow("Original Image", img)  
cv2.imshow("Rotated Image (90° Clockwise)", rotated)  
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

Output:



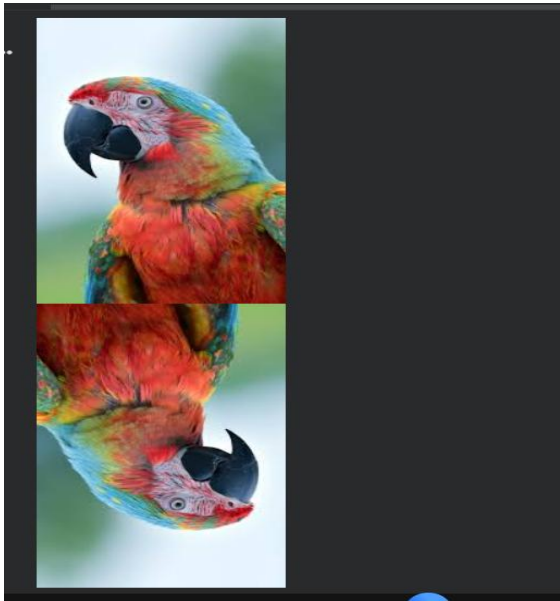
Exp-11:

180-degree rotation clockwise

Code:

```
import cv2  
img = cv2.imread("input.jpg")  
rotated = cv2.rotate(img, cv2.ROTATE_180)  
cv2.imshow("Original Image", img)  
cv2.imshow("Rotated Image (180°)", rotated)  
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

Output:



Exp-12:

270-degree rotation clockwise

Code:

```
import cv2
from google.colab import files
import matplotlib.pyplot as plt
path = list(files.upload().keys())[0]
img = cv2.cvtColor(cv2.imread(path), cv2.COLOR_BGR2RGB)
rotated = cv2.rotate(img, cv2.ROTATE_90_COUNTERCLOCKWISE)
plt.imshow(rotated)
plt.title("270° Clockwise Rotation")
plt.axis('off')
plt.show()
```

Output:



Exp-13:

270-degree rotation clockwise

Code:

```
import cv2
from google.colab import files
import matplotlib.pyplot as plt
import numpy as np
path = list(files.upload().keys())[0]
img = cv2.cvtColor(cv2.imread(path), cv2.COLOR_BGR2RGB)
rows, cols = img.shape[:2]
pts1 = np.float32([[50, 50], [200, 50], [50, 200]])
pts2 = np.float32([[10, 100], [200, 50], [100, 250]])
matrix = cv2.getAffineTransform(pts1, pts2)
transformed = cv2.warpAffine(img, matrix, (cols, rows))
plt.imshow(transformed)
plt.title("Affine Transformed Image")
plt.axis('off')
plt.show()
```

OUTPUT:



EXP-14

```
import cv2
from google.colab import files
import matplotlib.pyplot as plt
import numpy as np
path = list(files.upload().keys())[0]
img = cv2.cvtColor(cv2.imread(path), cv2.COLOR_BGR2RGB)
h, w = img.shape[:2]
pts1 = np.float32([[50,50], [w-50,50], [50,h-50], [w-50,h-50]])
pts2 = np.float32([[0,0], [w,0], [0,h], [w,h]])
matrix = cv2.getPerspectiveTransform(pts1, pts2)
output = cv2.warpPerspective(img, matrix, (w, h))
plt.imshow(output)
plt.title("Perspective Transformed Image")
plt.axis('off')
plt.show()
```

OUTPUT:



EXP-15

```
import cv2  
  
from google.colab import files  
import matplotlib.pyplot as plt  
path = list(files.upload().keys())[0]  
img = cv2.cvtColor(cv2.imread(path), cv2.COLOR_BGR2RGB)  
plt.imshow(img)  
plt.title("Original Image")  
plt.axis('off')  
plt.show()
```

OUTPUT:

